

Concavity

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9:18 PM

To utilize FOC we also need to argue that $V(k)$ is concave

Result 2: $V(k)$ is strictly concave if

- $T(k)$ - convex
- $F(k, k')$ - strictly concave

Proof: Use the same recursive approach

Suppose that V_0 is weakly concave:

$$V_0(\lambda k_1 + (1-\lambda)k_2) \geq \lambda V_0(k_1) + (1-\lambda)V_0(k_2)$$

Show that TV_0 is strictly concave:

$$(TV_0)(\lambda k_1 + (1-\lambda)k_2) > \lambda(TV_0)(k_1) + (1-\lambda)(TV_0)(k_2), \quad k_1 \neq k_2$$

Denote by $g(k)$ the optimal $k'(k)$ associated with V_0

Then

$$\begin{aligned} \lambda(TV_0)(k_1) + (1-\lambda)(TV_0)(k_2) &= \\ &= \lambda F(k_1, g(k_1)) + \lambda \beta V(g(k_1)) + (1-\lambda) F(k_2, g(k_2)) + (1-\lambda) \beta V(g(k_2)) \end{aligned}$$

< / F-strictly concave, V-concave /

$$< F(\lambda k_1 + (1-\lambda)k_2, \lambda g(k_1) + (1-\lambda)g(k_2)) + \beta V(\lambda g(k_1) + (1-\lambda)g(k_2))$$

$$\leq \text{ / if } \underbrace{\lambda g(k_1) + (1-\lambda)g(k_2)}_{(*)} \in \Gamma(\lambda k_1 + (1-\lambda)k_2) \text{ /}$$

$$\leq TV_0(\lambda k_1 + (1-\lambda)k_2) \quad (*)$$

When does $(*)$ hold?

$$\begin{aligned} T(k) \text{ - convex correspondence} &\Rightarrow \lambda g(k_1) + (1-\lambda)g(k_2) \in \Gamma(\lambda k_1 + (1-\lambda)k_2) \\ &\text{ since } g(k_1) \in T(k_1), \quad g(k_2) \in T(k_2) \end{aligned}$$

Example: $\Gamma(k) = \{k' : 0 \leq k' \leq f(k)\}$

$$\left. \begin{array}{l} k'_1 \leq f(k_1) \\ k'_2 \leq f(k_2) \end{array} \right\} \Rightarrow \lambda k'_1 + (1-\lambda)k'_2 \leq \lambda f(k'_1) + (1-\lambda)f(k'_2) \leq \text{ / f-concave / } \leq f(\lambda k'_1 + (1-\lambda)k'_2)$$

$$\left[\begin{array}{l} k_2' \leq f(k_2') \end{array} \right] \Rightarrow \dots \leq f\text{-concave} \leq f(2k_1' + (1-2)k_2')$$

Thus $T(k)$ is convex is f -concave

$\Rightarrow TV_0$ is strictly concave $\Rightarrow \lim_{n \rightarrow \infty}$, apply T once more

$\Rightarrow V(k)$ is strictly concave. $\square$

Can be verified that $u(\cdot), f(\cdot)$ satisfy the desired properties if $u(\cdot)$ - strictly concave, inc.
 $f(\cdot)$ - concave

Remark : Such recursive approach is very useful for characterizing $V(k)$:

- differentiability
- bounds
- comparative statics (how does $V(\cdot)$ change if a parameter changes).

Using the properties of the value function :

If the solution to (BE) is interior $\Rightarrow$ it satisfies FOC:

$$(k') : F_2(k, k') + \beta V'(k') = 0$$

Or in terms of $u(\cdot)$ and $f(\cdot)$

$$u'(f(k) - k') = \beta V'(k')$$

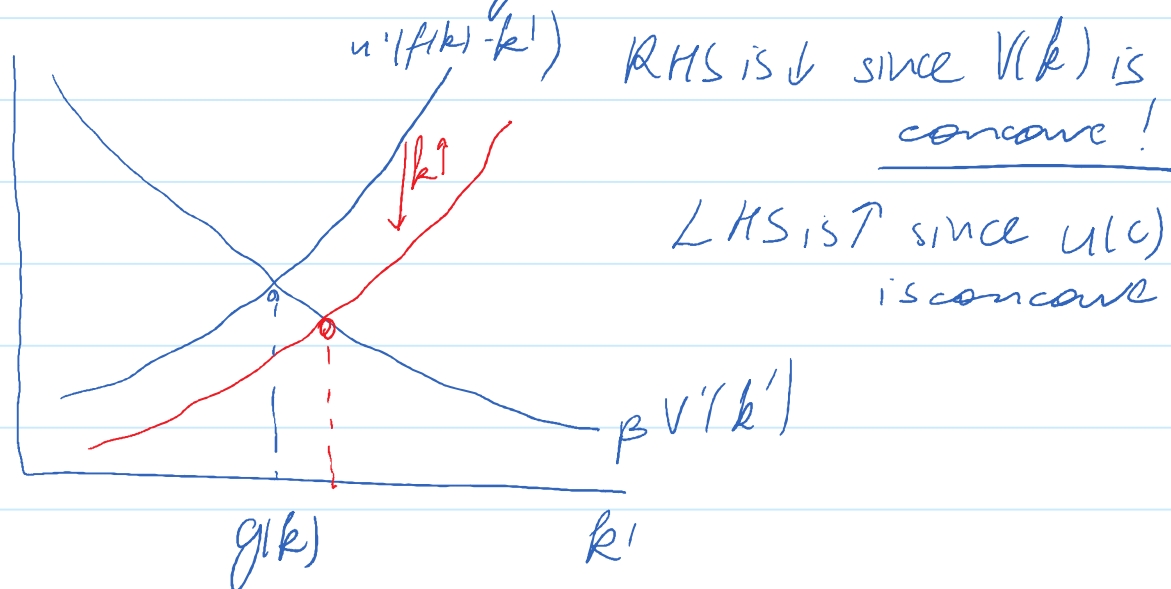
current cost = future benefit

(in terms of marginal utility)

(in terms of discounted future value)

Denote by $g(k)$ the optimal k'

What can we say about its properties?



$$k \uparrow \Rightarrow f(k) \uparrow \Rightarrow |u(c) - \text{concave}| \Rightarrow u'(f(k) - k') \downarrow$$

$R1 \Rightarrow f(k) \rightarrow |u(c)|\text{-concave} \Rightarrow u(rk/x)^v$

$\Rightarrow \boxed{g(k) \text{ is T in } k}$

Thus, concavity of $V(k)$ and $u(c)$ implies
that $g(k)$ is increasing
(save more if there is more
capital today)

Euler Equation -- once again!

$$V(k) = \max_{k' \in T(k)} \{ F(k, k') + \beta V(k') \}$$

If the solution is interior $\Rightarrow$

$$(FOC): \boxed{F_2(k, k') + \beta V'(k') = 0} \quad \Rightarrow \quad k' = g(k)$$

Envelope condition:

$$\begin{aligned} V(k) &= F(k, g(k)) + \beta V(g(k)) \\ \Rightarrow V'(k) &= F_1(k, g(k)) + \underbrace{[F_2(k, g(k)) + \beta V'(g(k))]}_{=0 \text{ by FOC}} g'(k) \end{aligned}$$

$$\Rightarrow \boxed{V'(k) = F_1(k, g(k))} \quad (EC) \text{ envelope condition}$$

$$(FOC + EC) \Rightarrow \boxed{F_2(k, g(k)) + \beta F_1(g(k), g(g(k))) = 0} \quad (EE)$$

familiar Euler Equation!!! ☺

Or rewrite it in terms of $u(\cdot)$ and $f(\cdot)$:

$$u'(f(k) - g(k)) = \beta u'(f(g(k)) - g(g(k))) \cdot f'(g(k))$$

Formally, this is a functional equation with respect to $g(k)$.

If a closed-form solution exists, it of course can be found using the guess-and-

verify method (guess $g(k)$, find coefficients).

However, (EE) does not hold if the solution is corner solution.

Or if the closed-form solution does not exist, (EE) might be hard to solve numerically: there is no "convergence" result like for $V(k)$ in the (BE) (contraction mapping)

Technically, one can rewrite (EE):

$$g(k) = \underbrace{f(k) - (u')^{-1}(p u'(f(g(k)) - g(g(k))) \cdot f'(g(k)))}_{= \text{operator on } g(k)}$$

But it is messy, likely not a contraction (w.r.t. which metrics?)

$\Rightarrow$ While (BE) is easy to solve numerically, (EE) may be tricky..